High Performance Computing Park (PCAD)

Parallel and Distributed Processing Group (GPPD-HPC)

Lucas Mello Schnorr

Institute of Informatics, UFRGS

E.A.T. Meeting
Administration Building, Institute of Physics, UFRGS
Porto Alegre, April 10, 2026, 14:00

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UFRGS UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL

History

Continuous evolving platform

- Heavily depends on faculty projects
  - Computational heterogeneity
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  - Students rotate through the system

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First general organization around ≈2018/2019

High Performance Computing Park (PCAD)

`https://pcad.inf.ufrgs.br/`

High Performance Computing Park (PCAD)

URL: `http://pcad.inf.ufrgs.br/`

Has approximately 50 nodes, 1000+ CPU cores and 100000+ GPU cores

Location: *Data Center* of the Institute of Informatics, UFRGS

Selected HW Configurations for Computation

Intel Xeon Processors

- E5530 Nehalem, X7550 Nehalem
- E5-2630 Sandy Bridge
- E5-2640 v2 Ivy Bridge
- E5-2650 v3 Haswell
- E5-2699 v4 Broadwell
- Gold (5317 Ice Lake, 6226 C. Lake)
- Silver (4208 C. Lake, 4116 Skylake)

Intel Core Processors

- Intel Core i7-10700F, i7-14700KF
- Intel Core i9-14900KF

NVIDIA Processors

- 2× NVIDIA Grace CPU Superchip

AMD Processors

- AMD Ryzen 9 3950X Zen2
- AMD RYZEN 5 3400G Zen+

NVIDIA Accelerators

- 6× GTX1080Ti
- 4× P100
- 1× RTX3090
- 5× RTX4070
- 6× RTX4090

Other Accelerators

- 4× NEC 10BE SX-Aurora
- 3× AMD Radeon RX 7900 XT

Desired Infrastructure Requirements

For Users

- To be independent in conducting experiments
- Easy to use (assuming basic Linux knowledge)
- Isolated, unrestricted, and maximum-performance access to resources (*bare metal*)
- Some conveniences (global file system, resource usage queue)

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For Administrators

- Mutualization and better utilization of computational resources
- Relatively easy installation $\rightarrow$ Low maintenance (ideally: *install and forget*)
- Logging of all user actions

General Platform Philosophy

- Warranty inspired by GPLv3, system provided “as-is”
  - No backups, no guarantee of operation, everything can be shut down without prior notice
  - Voluntary work by graduate students (PPGC)

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- Special requirements for experimental control
  - Control CPU and GPU frequency
  - Disable/enable cores, turboboost, hyperthreading, . . .
  - Specific BIOS configurations
  - Use of local disks (*scratch*) for data experiments

Base System and User/File Management

Base System → Debian GNU/Linux

- *Free Software*, has existed since ≈1993
- Stable version with LTS, currently Debian 12

User Management → LDAP (with OpenLDAP)

- *Free Software*, maintained by the OpenLDAP Foundation (since ≈1998)
  - Unification of all users and profile data (UID, GID)
- Access using SSH keys (no passwords)

Network File System → NFS

- System present directly in the Linux kernel
  - Transparent, does not require extra configuration by the user
- Shares the `$HOME` directory across all machines

Resource Sharing System → `slurm`

Slurm

- Schedule user allocations and tasks on machines
- Widely used in various supercomputers
  - Efficient (scheduling implementation and quality)
- Isolate environments (nodes or intra-node resources)

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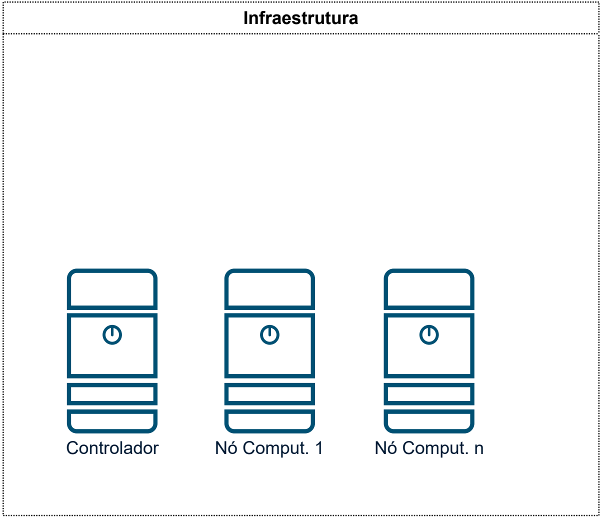
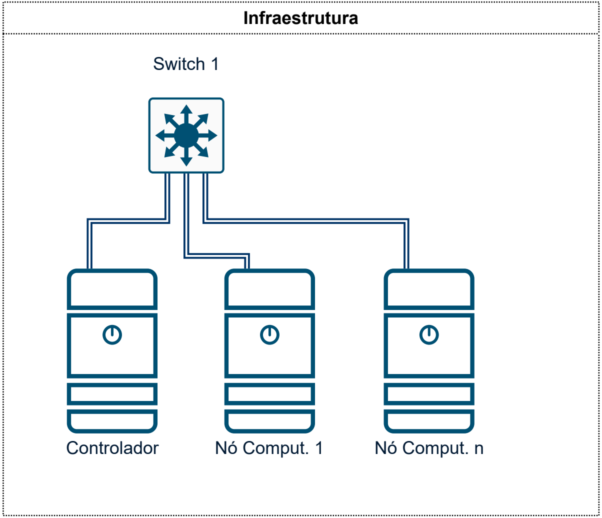
Slurm is a fundamental component for managing heterogeneity

- Uses the concept of partitions
  - Nodes belonging to a partition are homogeneous (almost always)
- We have $\approx$21 partitions, many with only 1 machine

Physical Interconnection Infrastructure



Physical Interconnection Infrastructure



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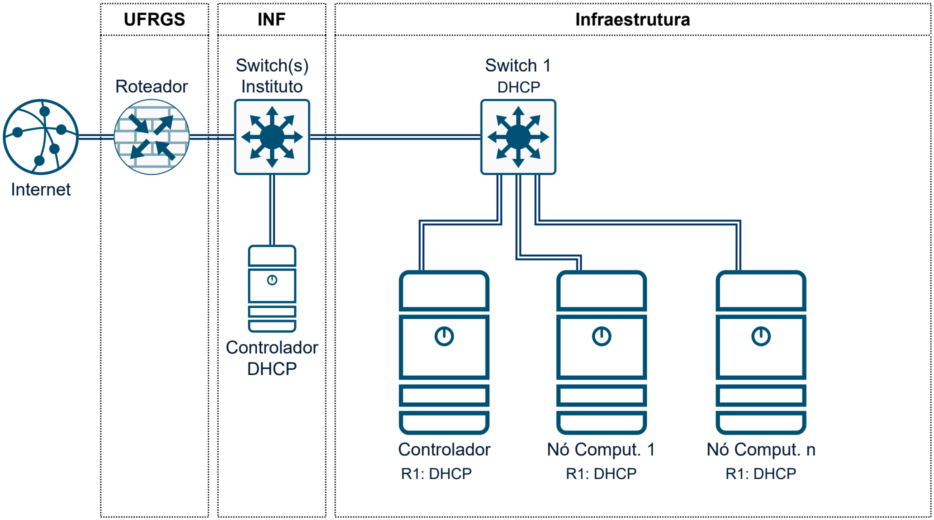

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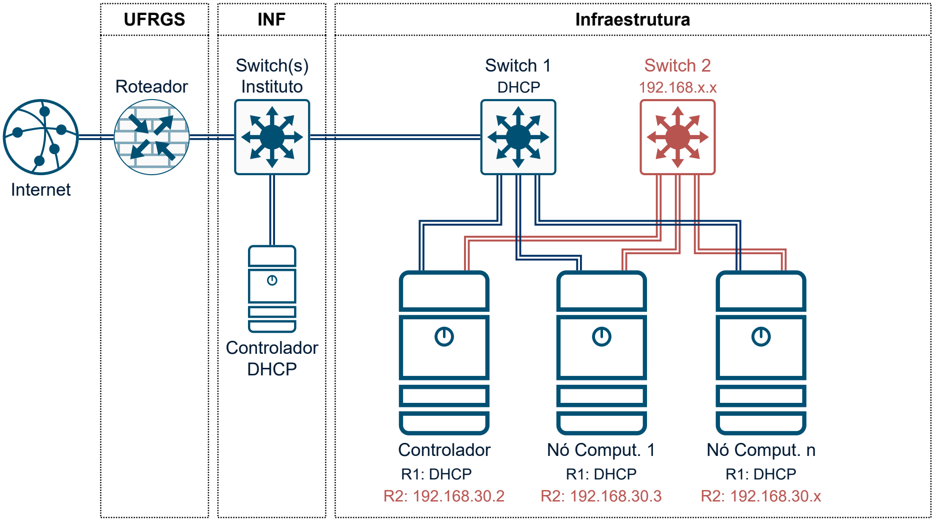


Upcoming Update 1/2 (560k USD)

We will receive via FINEP Proinfra Multi-Cad project an NVIDIA DGX B200 node

8x NVIDIA Blackwell GPUs (B200)
GPU RAM 1,440GB total
2x Intel Xeon Plat. 8570, 112 cores
RAM 2TB DDR5 4800 MT/s ECC Reg

4x OSFP, 2x QSFP112 (up to 400Gb/s)
2× 10GbE RJ45 (management)

OS Disk: 2x 1.9TB NVMe M.2
Internal Disk: 8x 3.84TB NVMe U.2

NVIDIA AI Enterprise
Mission Control + Run:ai
DGX OS (Ubuntu)

Rack units: 10Us - Weight: 142.4Kg
444mm (H) × 482.2mm (W) × 897.1mm (D)

NVIDIA DGX B200 Computational Performance

Main Computational Resource

8x NVIDIA Blackwell GPUs (B200)
GPU RAM 1,440GB total
DGX OS: CUDA, CUDA-X, NGC Catalog

FP Precision Comparison

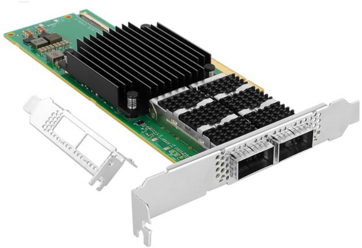
Precision	Per GPU (B200)	8x GPUs (DGX B200)
FP4	18 petaFLOPS	144 petaFLOPS
FP8	9 petaFLOPS	72 petaFLOPS
FP16 / BF16	4.5 petaFLOPS	~36 petaFLOPS
TF32	2.2 petaFLOPS	~17.6 petaFLOPS
FP32	75 teraFLOPS	~600 teraFLOPS
FP64	37 teraFLOPS	~296 teraFLOPS

Upcoming Update 2/2 (25k USD)

Virtually all compute nodes will be equipped with 100Gbit/s networking

Upgrade to 100Gbit/s Network

#	Qty	Description
01	8	ConnectX-4 Card 25Gb x8
02	11	ConnectX-4 Card 100Gb x16
03	11	ConnectX-6 Card 100Gb x16
04	1	ConnectX-6 Card 100Gb x8
05	2	100Gb Breakout Cable 4x25Gb 3m
06	2	100Gb Breakout Cable 4x25Gb 7m
07	12	100Gb Cable 3m
08	15	100Gb Cable 7m



And we will be connected to RNP's e-Science network via 100Gbit/s.

PCAD Access Demo

```
schnorr@larioja:~$ ssh pcad.inf.ufrgs.br
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 - PCAD Doc: http://gppd-hpc.inf.ufrgs.br/
 - Resource Monitor: http://gppd-hpc.inf.ufrgs.br/ganglia/
Last login: Fri Apr 10 07:02:33 2026 from 187.45.73.212
schnorr@gppd-hpc:~$
```

GPPD-HPC Research (`https://www.inf.ufrgs.br/gppd/site/`)

Three Faculty Members

- Philippe Olivier Alexandre Navaux
- Arthur Francisco Lorenzon
- Lucas Mello Schnorr

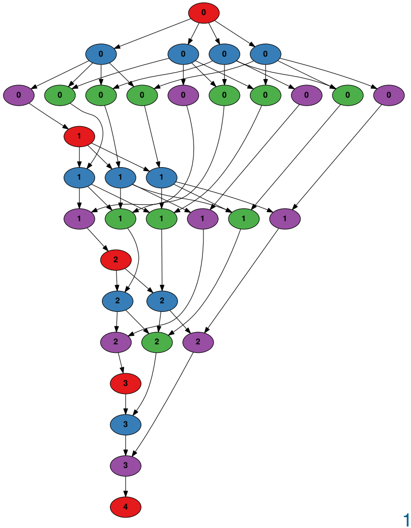
Cross-cutting Research Topics

Parallel programming	OpenMP, MPI, CUDA, task-based models (StarPU)
Scheduling	Load balancing, task/data mapping, DCT
Energy efficiency	Green computing, EDP, DVFS, frequency management at exascale
HPC Infrastructure	Cluster management (Slurm, BCM), PCAD, reproducibility
Scientific apps	Seismics, CFD, geophysical stencils, graphs, deep learning
HPC + AI	LLM training and inference, GPU sharing, AI Factories
Visualization	Performance traces, PajeNG tools, StarVZ
Simulation	Simulation of computational systems (SimGrid)

Research: Observing Behavior of Task-based Applications 1/2

Block Cholesky Factorization and its Directed-Acyclic Graph (for $N = 5$).

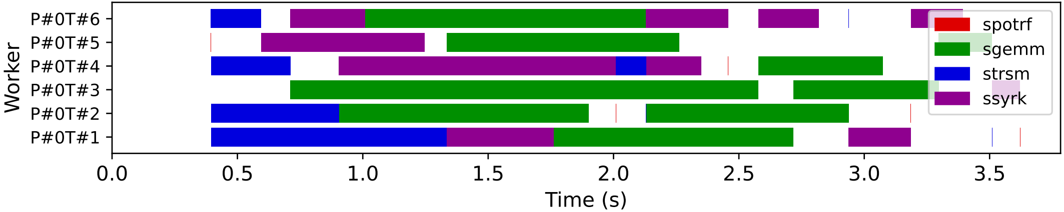
```
for (k = 0; k < N; k++) {
   DPOTRF(RW, A[k][k]);
   for (i = k+1; i < N; i++)
     DTRSM(RW, A[i][k], R, A[k][k]);
   for (i = k+1; i < N; i++) {
     DSYRK(RW, A[i][i], R, A[i][k]);
     for (j = k+1; j < i; j++)
        DGEMM(RW, A[i][j], R, A[i][k],
                           R, A[j][k]);
   }
}
```



Research: Observing Behavior of Task-based Applications 2/2

The StarPU *runtime* generates a *trace* of a DAG scheduling on resources.

R/StarVZ `https://cran.r-project.org/web/packages/starvz/` for analysis.



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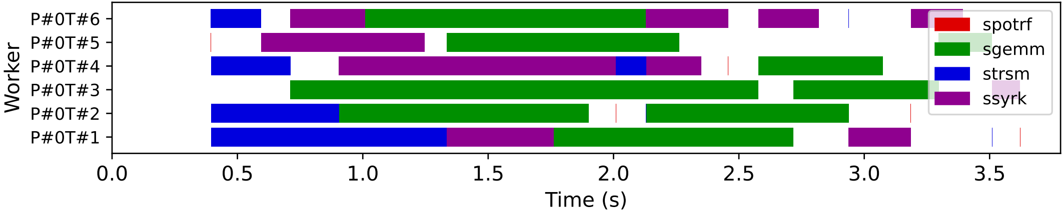

Typical Questions

- How to improve DAG scheduling?
- Given an application, how to arrive at a block algorithm?
- How to partition the DAG in heterogeneous systems at the system level?
- How to perform large-scale data visualization?

References

DevOps for HPC: How to Configure a Cluster for Shared Use

- `https://lnesi.gitlab.io/mc-hpc-share/`
- `https://cradrs.github.io/eradrs2023/pdfs/minicursos/cap-3.pdf`

Best Practices for HPC Experiments

- `https://exp-hpc.gitlab.io/`
- Series of short courses since 2019 at ERAD/RS, ERAD/SP, WSCAD

Parallel and Distributed Processing Group (GPPD/HPC)

- HPC, new architectures, parallel programming paradigms, performance analysis
- `https://www.inf.ufrgs.br/gppd/site/`

Contact

Thank you for your attention!

schnorr@inf.ufrgs.br

PCAD

`http://pcad.inf.ufrgs.br/`

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